DLNA/UPnP 技术调研

一，设计目标

1. 基于DLNA/UPnP，使得终端通过linux播放音频

二，DLNA/UPnP

1. 参考资料

a. https://blog.csdn.net/wilson1068/article/details/88574574

b. https://blog.csdn.net/u012910985/article/details/25866519

c. https://juejin.cn/post/7293786573677887515

d. https://blog.51cto.com/u_15940062/6008784

UPnP-arch-DeviceArchitecture-v1.1.pdf

UPnP_UDA_tutorial_July2014.pdf

2. 简介

a. DLNA（Digital Living Network Alliance，数字生活网络联盟）和 UPnP（Universal Plug and Play，通用即插即用）是两个相关的标准和协议，旨在实现设备之间的互联互通，共享多媒体内容的目标。

b. UPnP 是一个基于网络技术的框架和协议，旨在实现设备的即插即用和无缝互操作。UPnP 简化了设备的发现、连接和控制，使得各种设备可以自动识别和配置，实现设备之间的互联互通。DLNA 协议则是基于 UPnP 的特定应用场景的一个扩展。

综上所述，DLNA 是一个基于 DLNA 标准的多媒体内容共享和传输的解决方案，而 UPnP 是一个通用的设备互联和互操作框架，DLNA 是 UPnP 的一个具体应用扩展。通过使用 DLNA/UPnP，用户可以轻松实现设备间的音频、视频等多媒体内容共享和交互。

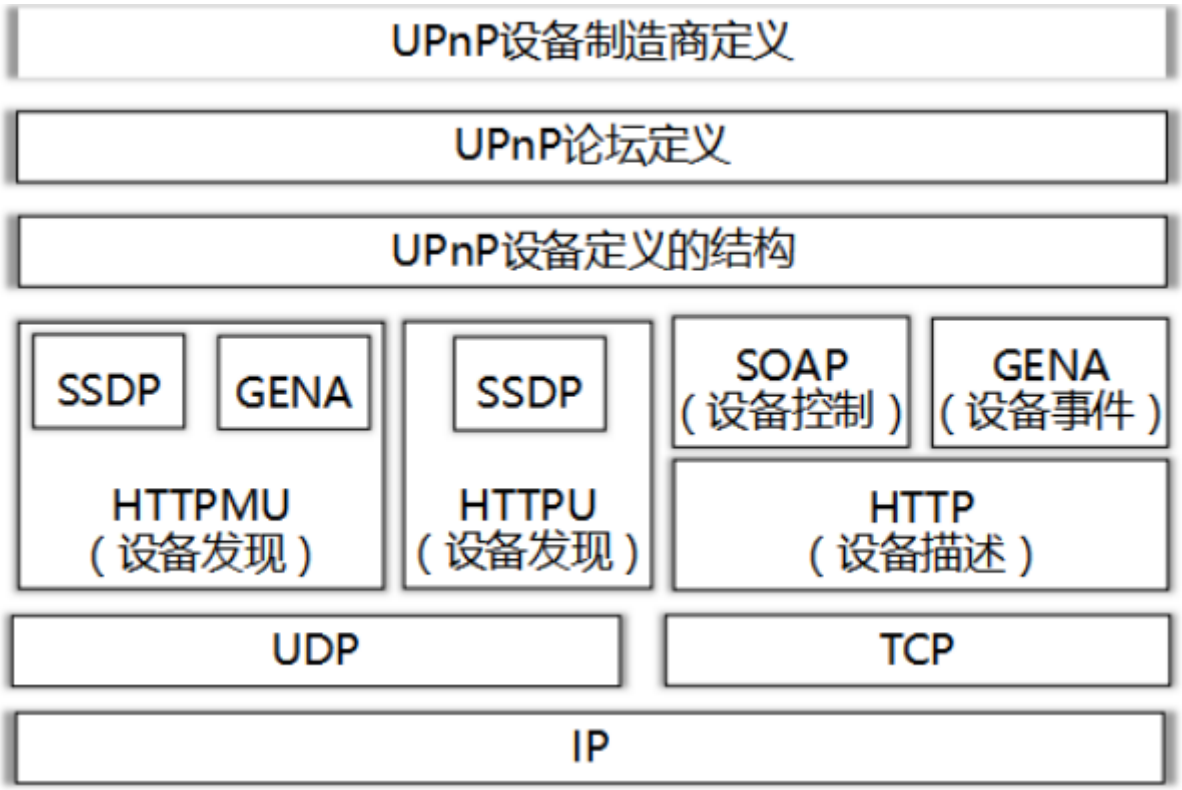
3. 关键组件

1. ControlPoint：是UPnP网络中的控制者，能发现和控制网络中的其他UPnP设备。控制点可以搜索网络中的upnp设备，获取设备的能力和服务，并与设备进行通信。它通过发送命令和请求来控制媒体服务器和媒体渲染器的行为。
2. MediaServer：是 UPnP 控制点的一个子模块，会开启流服务，共享音频媒体资源,并将流信息通过控制节点共享媒体渲染器进行播放。

3. MediaPlayCtrl：是 UPnP 控制点的一个子模块，负责控制媒体渲染器的播放状态。它可以发送命令来控制媒体渲染器的播放和暂停，调整音量、跳过轨道等
4. MediaRender：是 UPnP 网络中的一种设备，它可以接收媒体服务器发送的音频媒体流信息，并播放这些流。通过与控制点通信，媒体渲染器可以接受播放控制命令，如播放、暂停、停止、调节音量等，并将收到的媒体流进行播放。

4. 协议栈

UPnP是一个多层协议构成的框架体系，每一层都以相邻的下层为基础，同时又是相邻上层的基础。



- 第一，二层:TCP/IP和UDP ，UPnP是一种网络协议体系,UPnP设备之间的数据交换基于此协议
- 第三层是HTTP、HTTPU、HTTPMU，这一层属于传送协议层，传送的内容都是经过封装后的内容，存放在特定的XML文件中。对应的SSDP、SOAP、GENA指的是保存在XML文件中的数据格式。这一层，已经解决了UPnP设备的IP地址和传送信息问题；
- 第四层是UPnP设备体系定义，仅仅是一个抽象的、公用的设备模型。任何UPnP设备都必须使用这一层；
- 第五层是UPnP论坛的各个专业委员会的设备定义层，在这个论坛中，不同电器设备由不同的专业委员会定义
- 第六层，也就是应用层，由UPnP设备制造厂商定义的部分。这一层的信息是由设备制造厂商来“填充” 的，这部分一般有设备厂商提供的、对设备控制和操作的底层代码，然后，就是名称序列号呀，厂商信息之类的东西；

5. 协议分析

5.1 HTTP

HTTP是UPnP协议栈的核心协议之一，工作在TCP/IP之上，其中HTTPU和HTTPMU是HTTP协议的拓展，而SSDP、GENA、SOAP等协议则建立在HTTP的拓展协议之上

5.2 HTTPU和HTTPMU

HTTPU和HTTPMU中的“U”是UDP，采用非可靠的UDP/IP方式传送信息，协议格式基本与HTTP相同。HTTPMU的“M”是Multicast（广播)

1. HTTPMU 基于UDP的HTTP广播
2. HTTPU 基于UDP的HTTP单播

5.3 SSDP

SSDP是UPnP协议栈的核心协议之一。其建立在HTTPU和HTTPMU基础之上，通过广播发送信息以实现在局域网内发现设备或感兴趣的资源的一种应用层协议。控制点和设备都必须使用SSDP协议。当MediaRender加入网络后会广播发出通知信息NOTIFY, 当ControlPoint加入网络时会通过HTTPMU向特定的SSDP广播地址及端口发送搜索信息M-SEARCH。设备收到该广播的搜索信息时，将检查自身与搜索信息是否匹配，如果匹配将使用HTTPU进行响应，使控制点实现设备和服务的发现。

a. NOTIFY

GMediaRender启动后会广播通知消息,宣告UPnP设备可用性

- NOTIFY * HTTP/1.1\r\n ：请求头
- HOST：广播地址端口
- LOCATION：设备描述url
- NT：设备类型
- NTS：通知消息子设备类型
- USN：设备标识符

i. **upnp:rootdevice**

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n  #设备描述文件的URL
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: ace28da8-dc40-11ee-a330-ff919167c84f\r\n #设备标识符
    NT: upnp:rootdevice\r\n #设备类型
    NTS: ssdp:alive\r\n    #通知消息消息子类型
```

```
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002::upnp:rootdevice\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]

```

ii. uuid:GMediaRender-1_0-000-000-002

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: ace28da8-dc40-11ee-a330-ff919167c84f\r\n
    NT: uuid:GMediaRender-1_0-000-000-002\r\n #子设备
    NTS: ssdp:alive\r\n
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]
```

iii. **urn:schemas-upnp-org:device:MediaRenderer:1**

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: ace28da8-dc40-11ee-a330-ff919167c84f\r\n
    NT: urn:schemas-upnp-org:device:MediaRenderer:1\r\n
    NTS: ssdp:alive\r\n
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:device:MediaRenderer:1\r\n
    \r\n
```

```
    [Full request URI: http://239.255.255.250:1900*]
```

iv. urn:schemas-upnp-org:service:AVTransport:1\r\n

音视频流的传输控制,用于控制活动和位置，比如播放、暂定、快进、快退等;

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: ace28da8-dc40-11ee-a330-ff919167c84f\r\n
    NT: urn:schemas-upnp-org:service:AVTransport:1\r\n
    NTS: ssdp:alive\r\n
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:service:AVTransport:1\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]
```

v. urn:schemas-upnp-org:service:ConnectionManager:1

用于列举支持的传输协议和媒体格式,控制点与自身的连接管理

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: ace28da8-dc40-11ee-a330-ff919167c84f\r\n
    NT: urn:schemas-upnp-org:service:ConnectionManager:1\r\n
    NTS: ssdp:alive\r\n
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:service:ConnectionManager:1\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]
```

vi. **urn:schemas-upnp-org:service:RenderingControl:1**

支持控制点发现设备所支持的状态变量表,获取当前任何变量状态,改变可修改的状态变量,初始化设置，比如音量等

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: ace28da8-dc40-11ee-a330-ff919167c84f\r\n
    NT: urn:schemas-upnp-org:service:RenderingControl:1\r\n
    NTS: ssdp:alive\r\n
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:service:RenderingControl:1\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]
```

i. 退出

```
Simple Service Discovery Protocol
    NOTIFY * HTTP/1.1\r\n
    HOST: 239.255.255.250:1900\r\n
    CACHE-CONTROL: max-age=100\r\n
    LOCATION: http://192.168.50.44:49494/description.xml\r\n
    OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01\r\n
    01-NLS: e18bd04a-dc4b-11ee-ab84-f227fad14943\r\n
    NT: urn:schemas-upnp-org:service:RenderingControl:1\r\n
    NTS: ssdp:byebye\r\n
    SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4\r\n
    X-User-Agent: redsonic\r\n
    USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:service:RenderingControl:1\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]

```

b. M-SEARCH

i. 控制点(ControlPoint)发现可用UPnP设备

CP使用UDP多播包向组播地址发送HTTP的M-SEARCH命令，任何网络上服务控制点搜索条件的设备发回一个UDP单播进行应答,该应答中包含了指向其描述文档的 URL 地址。

- MAN: "ssdp:discover" 标准规定
- ST：服务查询的目标；

  ssdp:all：搜索所有设备和服务

  upnp:rootdevice：仅搜索网络中的根设备

  uuid:device-UUID：查询UUID标识的设备

  urn:schemas-upnp-org:device:device-Type:version：查询device-Type字段指定的设备类型，设备类型和版本由UPNP组织定义

  urn:schemas-upnp-org:service:service-Type:version：查询service-Type字段指定的服务类型，服务类型和版本由UPNP组织定义

```
# andorid
Internet Protocol Version 4, Src: 192.168.50.84, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 40972, Dst Port: 1900
Simple Service Discovery Protocol
    M-SEARCH * HTTP/1.1\r\n
    ST: urn:schemas-upnp-org:device:MediaRenderer:1\r\n
    HOST: 239.255.255.250:1900\r\n
    MX: 10\r\n
    MAN: "ssdp:discover"\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]
    [HTTP request 10/25]
    [Prev request in frame: 134]
    [Next request in frame: 175]
```

ii. 应答

```
Internet Protocol Version 4, Src: 192.168.50.44, Dst: 192.168.50.84
User Datagram Protocol, Src Port: 58426, Dst Port: 40972

HTTP/1.1 200 OK
CACHE-CONTROL: max-age=100
DATE: Thu, 07 Mar 2024 05:49:00 GMT
EXT:
LOCATION: http://192.168.50.44:49494/description.xml #返回表述信息
OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01
```

```
01-NLS: a071a360-dc45-11ee-9d0a-fea9db7b1101
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic
ST: urn:schemas-upnp-org:device:MediaRenderer:1
USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:device:MediaRenderer:1
```

5.4 GENA

GENA以HTTP协议为基础，为实现在HTTP资源之间发送和接受事件通知而建立的一个基于HTTP的通知架构。GENA定义了事件通知过程中的操作对象：提供事件，并在事件发生时发送事件通知的事件发布者（publisher）；订阅事件，接收事件通知的事件订阅者（subscriber）。

事件通知有两种形式：

- 单播事件。事件发生变化后，事件发布者向订阅了该事件的事件订阅者单独发送通知，采用可靠的TCP/IP发送机制。在UPnP协议栈中，采用HTTP协议。
- 广播事件。事件发生变化后，事件发布者向所有订阅了该事件的事件订阅者广播发送通知，采用非可靠的UDP广播发送。在UPnP协议栈中，采用HTTPMU协议

5.5 SOAP

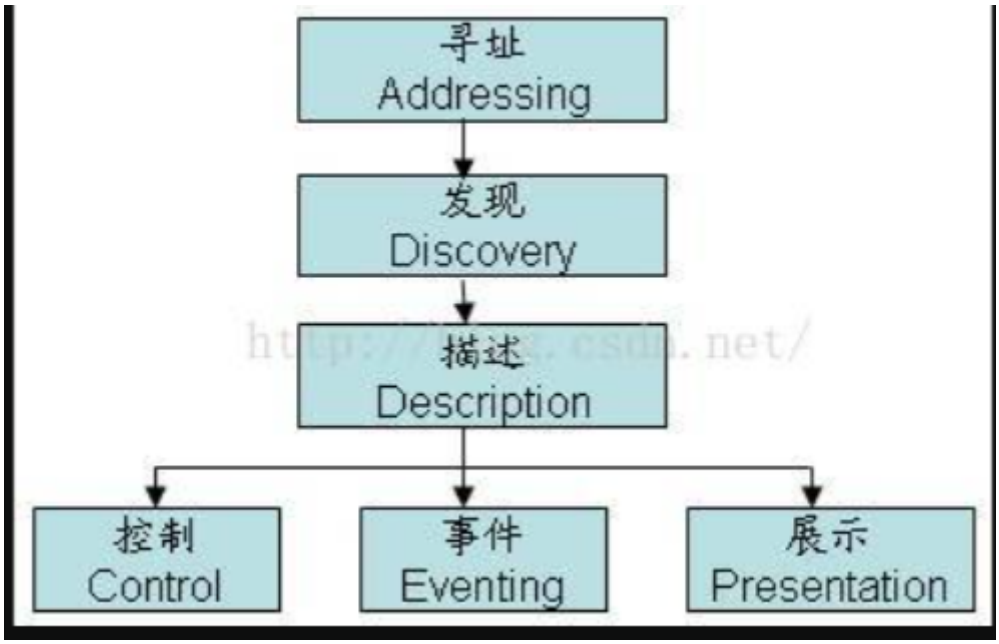
SOAP基于HTTP和XML协议，用于实现UPnP网络内的控制点和设备之间控制信息的传递。控制点向设备发送的所有控制请求都使用SOAP封装，而设备收到SOAP请求在执行完毕后将返回执行结果或者出错信息的过程采用SOAP响应。

5.6 XML

XML是UPnP协议体系中很重要的组成，被用于描述（设备描述和服务描述），控制消息和事件通知中。

6. 工作流程

流程图



6.1 寻址

6.2 发现

ControlPoint与MediaRender发现彼此并建立通信,基于SSDP协议实现，

a. ControlPoint

i. 宣告自己的存在

39	25.323701	192.168.50.84	239.255.255.250	SSDP	361 NOTIFY * HTTP/1.1
41	25.325157	192.168.50.84	239.255.255.250	SSDP	398 NOTIFY * HTTP/1.1
43	25.328143	192.168.50.84	239.255.255.250	SSDP	400 NOTIFY * HTTP/1.1
45	25.330862	192.168.50.84	239.255.255.250	SSDP	356 NOTIFY * HTTP/1.1
47	25.331826	192.168.50.84	239.255.255.250	SSDP	365 NOTIFY * HTTP/1.1
49	25.332676	192.168.50.84	239.255.255.250	SSDP	402 NOTIFY * HTTP/1.1
51	25.334304	192.168.50.84	239.255.255.250	SSDP	404 NOTIFY * HTTP/1.1

ii. 查看可用设备

	Time	Source	Destination	Protocol	Length Info
19	22.527891	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1
22	23.030146	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1
25	23.533681	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1
27	24.037100	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1
30	24.414266	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1
32	24.539110	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1
34	24.917622	192.168.50.84	239.255.255.250	SSDP	174 M-SEARCH * HTTP/1.1

b. **MediaRender**

i. 宣告自己的存在

60728	704.339786218	192.168.50.1	239.255.255.250	SSDP	521 NOTIFY * HTTP/1.1
60729	704.339840156	192.168.50.1	239.255.255.250	SSDP	503 NOTIFY * HTTP/1.1
60730	704.339897464	192.168.50.1	239.255.255.250	SSDP	450 NOTIFY * HTTP/1.1
60731	704.339956657	192.168.50.1	239.255.255.250	SSDP	489 NOTIFY * HTTP/1.1
60732	704.340022668	192.168.50.1	239.255.255.250	SSDP	450 NOTIFY * HTTP/1.1
60733	704.340090218	192.168.50.1	239.255.255.250	SSDP	509 NOTIFY * HTTP/1.1
60734	704.340153346	192.168.50.1	239.255.255.250	SSDP	450 NOTIFY * HTTP/1.1
60735	704.340219055	192.168.50.1	239.255.255.250	SSDP	513 NOTIFY * HTTP/1.1

c. **控制点(ControlPoint)发现可用UPnP设备(MediaRender)**

ControlPoint：192.168.50.84

MediaRender：192.168.50.44

ControlPoint广播查找可用UPnP设备(MediaRender),MediaRender收到消息后应答

1403	8.055768771	192.168.50.84	239.255.255.250	SSDP	172	M-SEARCH * HTTP/1.1
1404	8.055996039	192.168.50.44	192.168.50.84	UDP	518	58916 → 50621 Len=476

i. 查找

```
# andorid
Internet Protocol Version 4, Src: 192.168.50.84, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 40972, Dst Port: 1900
Simple Service Discovery Protocol
    M-SEARCH * HTTP/1.1\r\n
    ST: urn:schemas-upnp-org:device:MediaRenderer:1\r\n
    HOST: 239.255.255.250:1900\r\n
    MX: 10\r\n
    MAN: "ssdp:discover"\r\n
    \r\n
    [Full request URI: http://239.255.255.250:1900*]
    [HTTP request 10/25]
```

ii. 应答

1. 核心内容:http://192.168.50.44:49494/description.xml

```
Internet Protocol Version 4, Src: 192.168.50.44, Dst: 192.168.50.84
User Datagram Protocol, Src Port: 58426, Dst Port: 40972

HTTP/1.1 200 OK
CACHE-CONTROL: max-age=100
DATE: Thu, 07 Mar 2024 05:49:00 GMT
EXT:
LOCATION: http://192.168.50.44:49494/description.xml
OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01
01-NLS: a071a360-dc45-11ee-9d0a-fea9db7b1101
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic
ST: urn:schemas-upnp-org:device:MediaRenderer:1
USN: uuid:GMediaRender-1_0-000-000-002::urn:schemas-upnp-org:device:MediaRenderer:1
```

6.3 描述

当控制点发现一个可用设备后，需要获取更加详细的信息，因而会向UPnP设备请求设备描述文档。描述文档是一个XML文档，用来描述一个设备。

6.3.1 description.xml

```
GET /description.xml HTTP/1.1
Accept-Encoding: gzip
User-Agent: Jetty/9.3.z-SNAPSHOT
Host: 192.168.50.44:49494

HTTP/1.1 200 OK
CONTENT-LENGTH: 2052
Accept-Ranges: bytes
CONTENT-TYPE: text/xml; charset="utf-8"
DATE: Thu, 07 Mar 2024 07:15:37 GMT
LAST-MODIFIED: Thu, 07 Mar 2024 07:15:29 GMT
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic
CONNECTION: close
```

```
<root xmlns="urn:schemas-upnp-org:device-1-0">
    <specVersion>
        <major>1</major>
        <minor>0</minor>
    </specVersion>
    <device> //deviceType 设备类型
        <deviceType>urn:schemas-upnp-org:device:MediaRenderer:1</deviceType>
        <presentationURL/>
        <friendlyName>note_note</friendlyName> //设备名称
        <manufacturer>Ivo Clarysse, Henner Zeller</manufacturer>
        <manufacturerURL>http://github.com/hzeller/gmrender-resurrect</manufacturerURL>
        <modelDescription>gmediarender 0.0.9</modelDescription>
        <modelName>gmediarender</modelName>
        <modelNumber>0.0.9</modelNumber>
        <modelURL>http://github.com/hzeller/gmrender-resurrect</modelURL>
        <UDN>uuid:GMediaRender-1_0-000-000-002</UDN> //uuid
        <iconList>
            <icon>
                <mimetype>image/png</mimetype>
                <width>64</width>
                <height>64</height>
```

```
                <depth>24</depth>
                <url>/upnp/grender-64x64.png</url>
            </icon>
            <icon>
                <mimetype>image/png</mimetype>
                <width>128</width>
                <height>128</height>
                <depth>24</depth>
                <url>/upnp/grender-128x128.png</url>
            </icon>
        </iconList>
        <serviceList>  //服务列表
            <service>
                <serviceType>urn:schemas-upnp-org:service:AVTransport:1</serviceType> //服务类型
                <serviceId>urn:upnp-org:serviceId:AVTransport</serviceId>
                <SCPDURL>/upnp/rendertransportSCPD.xml</SCPDURL>  //服务描述url
                <controlURL>/upnp/control/rendertransport1</controlURL> //控制url
                <eventSubURL>/upnp/event/rendertransport1</eventSubURL> //事件订阅url
            </service>
            <service>
                <serviceType>urn:schemas-upnp-org:service:ConnectionManager:1</serviceType>
                <serviceId>urn:upnp-org:serviceId:ConnectionManager</serviceId>
                <SCPDURL>/upnp/renderconnmgrSCPD.xml</SCPDURL>
                <controlURL>/upnp/control/renderconnmgr1</controlURL>
                <eventSubURL>/upnp/event/renderconnmgr1</eventSubURL>
            </service>
            <service>
                <serviceType>urn:schemas-upnp-org:service:RenderingControl:1</serviceType>
                <serviceId>urn:upnp-org:serviceId:RenderingControl</serviceId>
                <SCPDURL>/upnp/rendercontrolSCPD.xml</SCPDURL>
                <controlURL>/upnp/control/rendercontrol1</controlURL>
                <eventSubURL>/upnp/event/rendercontrol1</eventSubURL>
            </service>
        </serviceList>
    </device>
    <URLBase>http://192.168.50.44:49494/</URLBase>
</root>
```

6.3.2 urn:schemas-upnp-org:service:AVTransport:1

在该服务描述文档中，包含了service支持的action指令，例如play、stop、setAVTransportURI、next以及previous等常见的一些指令，以及对应事件响应的定义，主要包括状态变量的变化情况

```
GET /upnp/rendertransportSCPD.xml HTTP/1.1
Accept-Encoding: gzip
User-Agent: Jetty/9.3.z-SNAPSHOT
Host: 192.168.50.44:49494

HTTP/1.1 200 OK
CONTENT-LENGTH: 15697
Accept-Ranges: bytes
CONTENT-TYPE: text/xml
DATE: Thu, 07 Mar 2024 07:15:37 GMT
LAST-MODIFIED: Thu, 01 Jan 1970 00:00:00 GMT
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic
CONNECTION: close
```

```
<sites>
    <site>
        <name>RUNOOB</name>
        <url>www.runoob.com</url>
    </site>This XML file does not appear to have any style information associated with it. The document tree is shown below.

    <scpd
        xmlns="urn:schemas-upnp-org:service-1-0">
        <specVersion>
            <major>1</major>
            <minor>0</minor>
        </specVersion>
        <actionList>
            <action>
                <name>GetCurrentTransportActions</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>Actions</name>
```

```
                        <direction>out</direction>
<relatedStateVariable>CurrentTransportActions</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>GetDeviceCapabilities</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>PlayMedia</name>
                        <direction>out</direction>
<relatedStateVariable>PossiblePlaybackStorageMedia</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>RecMedia</name>
                        <direction>out</direction>
<relatedStateVariable>PossibleRecordStorageMedia</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>RecQualityModes</name>
                        <direction>out</direction>
<relatedStateVariable>PossibleRecordQualityModes</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>GetMediaInfo</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>NrTracks</name>
                        <direction>out</direction>
```

```
<relatedStateVariable>NumberOfTracks</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>MediaDuration</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentMediaDuration</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentURI</name>
                        <direction>out</direction>
<relatedStateVariable>AVTransportURI</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentURIMetaData</name>
                        <direction>out</direction>
<relatedStateVariable>AVTransportURIMetaData</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>NextURI</name>
                        <direction>out</direction>
<relatedStateVariable>NextAVTransportURI</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>NextURIMetaData</name>
                        <direction>out</direction>
<relatedStateVariable>NextAVTransportURIMetaData</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>PlayMedium</name>
                        <direction>out</direction>
<relatedStateVariable>PlaybackStorageMedium</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>RecordMedium</name>
                        <direction>out</direction>
<relatedStateVariable>RecordStorageMedium</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>WriteStatus</name>
```

```
                        <direction>out</direction>
<relatedStateVariable>RecordMediumWriteStatus</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>GetPositionInfo</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>Track</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentTrack</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>TrackDuration</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentTrackDuration</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>TrackMetaData</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentTrackMetaData</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>TrackURI</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentTrackURI</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>RelTime</name>
                        <direction>out</direction>
<relatedStateVariable>RelativeTimePosition</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>AbsTime</name>
```

```
                        <direction>out</direction>
                        <relatedStateVariable>AbsoluteTimePosition</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>RelCount</name>
                        <direction>out</direction>
                        <relatedStateVariable>RelativeCounterPosition</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>AbsCount</name>
                        <direction>out</direction>
                        <relatedStateVariable>AbsoluteCounterPosition</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>GetTransportInfo</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
                        <relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentTransportState</name>
                        <direction>out</direction>
                        <relatedStateVariable>TransportState</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentTransportStatus</name>
                        <direction>out</direction>
                        <relatedStateVariable>TransportStatus</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentSpeed</name>
                        <direction>out</direction>
                        <relatedStateVariable>TransportPlaySpeed</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
```

```
            <action>
                <name>GetTransportSettings</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>PlayMode</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentPlayMode</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>RecQualityMode</name>
                        <direction>out</direction>
<relatedStateVariable>CurrentRecordQualityMode</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>Pause</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>Play</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>Speed</name>
                        <direction>in</direction>
```

```
<relatedStateVariable>TransportPlaySpeed</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>Seek</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>Unit</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_SeekMode</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>Target</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_SeekTarget</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>SetAVTransportURI</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentURI</name>
                        <direction>in</direction>
<relatedStateVariable>AVTransportURI</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>CurrentURIMetaData</name>
                        <direction>in</direction>
```

```
<relatedStateVariable>AVTransportURIMetaData</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>Stop</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
            <action>
                <name>SetNextAVTransportURI</name>
                <argumentList>
                    <argument>
                        <name>InstanceID</name>
                        <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>NextURI</name>
                        <direction>in</direction>
<relatedStateVariable>NextAVTransportURI</relatedStateVariable>
                    </argument>
                    <argument>
                        <name>NextURIMetaData</name>
                        <direction>in</direction>
<relatedStateVariable>NextAVTransportURIMetaData</relatedStateVariable>
                    </argument>
                </argumentList>
            </action>
        </actionList>
        <serviceStateTable>
            <stateVariable sendEvents="no">
                <name>TransportStatus</name>
                <dataType>string</dataType>
                <allowedValueList>
                    <allowedValue>OK</allowedValue>
                    <allowedValue>ERROR_OCCURRED</allowedValue>
```

```
307                     <allowedValue> vendor-defined </allowedValue>
308                 </allowedValueList>
309             </stateVariable>
310             <stateVariable sendEvents="no">
311                 <name>NextAVTransportURI</name>
312                 <dataType>string</dataType>
313             </stateVariable>
314             <stateVariable sendEvents="no">
315                 <name>NextAVTransportURIMetaData</name>
316                 <dataType>string</dataType>
317             </stateVariable>
318             <stateVariable sendEvents="no">
319                 <name>CurrentTrackMetaData</name>
320                 <dataType>string</dataType>
321             </stateVariable>
322             <stateVariable sendEvents="no">
323                 <name>RelativeCounterPosition</name>
324                 <dataType>i4</dataType>
325             </stateVariable>
326             <stateVariable sendEvents="no">
327                 <name>A_ARG_TYPE_InstanceID</name>
328                 <dataType>ui4</dataType>
329             </stateVariable>
330             <stateVariable sendEvents="no">
331                 <name>A_ARG_TYPE_SeekTarget</name>
332                 <dataType>string</dataType>
333             </stateVariable>
334             <stateVariable sendEvents="no">
335                 <name>PlaybackStorageMedium</name>
336                 <dataType>string</dataType>
337                 <allowedValueList>
338                     <allowedValue>UNKNOWN</allowedValue>
339                     <allowedValue>DV</allowedValue>
340                     <allowedValue>MINI-DV</allowedValue>
341                     <allowedValue>VHS</allowedValue>
342                     <allowedValue>W-VHS</allowedValue>
343                     <allowedValue>S-VHS</allowedValue>
344                     <allowedValue>D-VHS</allowedValue>
345                     <allowedValue>VHSC</allowedValue>
346                     <allowedValue>VIDEO8</allowedValue>
347                     <allowedValue>HI8</allowedValue>
348                     <allowedValue>CD-ROM</allowedValue>
349                     <allowedValue>CD-DA</allowedValue>
350                     <allowedValue>CD-R</allowedValue>
351                     <allowedValue>CD-RW</allowedValue>
352                     <allowedValue>VIDEO-CD</allowedValue>
353                     <allowedValue>SACD</allowedValue>
```

```
354                     <allowedValue>MD-AUDIO</allowedValue>
355                     <allowedValue>MD-PICTURE</allowedValue>
356                     <allowedValue>DVD-ROM</allowedValue>
357                     <allowedValue>DVD-VIDEO</allowedValue>
358                     <allowedValue>DVD-R</allowedValue>
359                     <allowedValue>DVD+RW</allowedValue>
360                     <allowedValue>DVD-RW</allowedValue>
361                     <allowedValue>DVD-RAM</allowedValue>
362                     <allowedValue>DVD-AUDIO</allowedValue>
363                     <allowedValue>DAT</allowedValue>
364                     <allowedValue>LD</allowedValue>
365                     <allowedValue>HDD</allowedValue>
366                     <allowedValue>MICRO-MV</allowedValue>
367                     <allowedValue>NETWORK</allowedValue>
368                     <allowedValue>NONE</allowedValue>
369                     <allowedValue>NOT_IMPLEMENTED</allowedValue>
370                     <allowedValue> vendor-defined </allowedValue>
371                 </allowedValueList>
372             </stateVariable>
373             <stateVariable sendEvents="no">
374                 <name>RelativeTimePosition</name>
375                 <dataType>string</dataType>
376             </stateVariable>
377             <stateVariable sendEvents="no">
378                 <name>PossibleRecordStorageMedia</name>
379                 <dataType>string</dataType>
380             </stateVariable>
381             <stateVariable sendEvents="no">
382                 <name>CurrentPlayMode</name>
383                 <dataType>string</dataType>
384                 <allowedValueList>
385                     <allowedValue>NORMAL</allowedValue>
386                     <allowedValue>REPEAT_ALL</allowedValue>
387                     <allowedValue>INTRO</allowedValue>
388                 </allowedValueList>
389                 <defaultValue>NORMAL</defaultValue>
390             </stateVariable>
391             <stateVariable sendEvents="no">
392                 <name>TransportPlaySpeed</name>
393                 <dataType>string</dataType>
394                 <allowedValueList>
395                     <allowedValue>1</allowedValue>
396                     <allowedValue> vendor-defined </allowedValue>
397                 </allowedValueList>
398             </stateVariable>
399             <stateVariable sendEvents="no">
400                 <name>PossiblePlaybackStorageMedia</name>
```

```
401                 <dataType>string</dataType>
402             </stateVariable>
403             <stateVariable sendEvents="no">
404                 <name>AbsoluteTimePosition</name>
405                 <dataType>string</dataType>
406             </stateVariable>
407             <stateVariable sendEvents="no">
408                 <name>CurrentTrack</name>
409                 <dataType>ui4</dataType>
410                 <allowedValueRange>
411                     <minimum>0</minimum>
412                     <maximum>4294967295</maximum>
413                     <step>1</step>
414                 </allowedValueRange>
415             </stateVariable>
416             <stateVariable sendEvents="no">
417                 <name>CurrentTrackURI</name>
418                 <dataType>string</dataType>
419             </stateVariable>
420             <stateVariable sendEvents="no">
421                 <name>CurrentTransportActions</name>
422                 <dataType>string</dataType>
423             </stateVariable>
424             <stateVariable sendEvents="no">
425                 <name>NumberOfTracks</name>
426                 <dataType>ui4</dataType>
427                 <allowedValueRange>
428                     <minimum>0</minimum>
429                     <maximum>4294967295</maximum>
430                 </allowedValueRange>
431             </stateVariable>
432             <stateVariable sendEvents="no">
433                 <name>AVTransportURI</name>
434                 <dataType>string</dataType>
435             </stateVariable>
436             <stateVariable sendEvents="no">
437                 <name>AbsoluteCounterPosition</name>
438                 <dataType>i4</dataType>
439             </stateVariable>
440             <stateVariable sendEvents="no">
441                 <name>CurrentRecordQualityMode</name>
442                 <dataType>string</dataType>
443                 <allowedValueList>
444                     <allowedValue>0:EP</allowedValue>
445                     <allowedValue>1:LP</allowedValue>
446                     <allowedValue>2:SP</allowedValue>
447                     <allowedValue>0:BASIC</allowedValue>
```

```
448                     <allowedValue>1:MEDIUM</allowedValue>
449                     <allowedValue>2:HIGH</allowedValue>
450                     <allowedValue>NOT_IMPLEMENTED</allowedValue>
451                     <allowedValue> vendor-defined </allowedValue>
452                 </allowedValueList>
453             </stateVariable>
454             <stateVariable sendEvents="no">
455                 <name>CurrentMediaDuration</name>
456                 <dataType>string</dataType>
457             </stateVariable>
458             <stateVariable sendEvents="no">
459                 <name>A_ARG_TYPE_SeekMode</name>
460                 <dataType>string</dataType>
461                 <allowedValueList>
462                     <allowedValue>ABS_TIME</allowedValue>
463                     <allowedValue>REL_TIME</allowedValue>
464                     <allowedValue>ABS_COUNT</allowedValue>
465                     <allowedValue>REL_COUNT</allowedValue>
466                     <allowedValue>TRACK_NR</allowedValue>
467                     <allowedValue>CHANNEL_FREQ</allowedValue>
468                     <allowedValue>TAPE-INDEX</allowedValue>
469                     <allowedValue>FRAME</allowedValue>
470                 </allowedValueList>
471             </stateVariable>
472             <stateVariable sendEvents="no">
473                 <name>AVTransportURIMetaData</name>
474                 <dataType>string</dataType>
475             </stateVariable>
476             <stateVariable sendEvents="no">
477                 <name>RecordStorageMedium</name>
478                 <dataType>string</dataType>
479                 <allowedValueList>
480                     <allowedValue>UNKNOWN</allowedValue>
481                     <allowedValue>DV</allowedValue>
482                     <allowedValue>MINI-DV</allowedValue>
483                     <allowedValue>VHS</allowedValue>
484                     <allowedValue>W-VHS</allowedValue>
485                     <allowedValue>S-VHS</allowedValue>
486                     <allowedValue>D-VHS</allowedValue>
487                     <allowedValue>VHSC</allowedValue>
488                     <allowedValue>VIDEO8</allowedValue>
489                     <allowedValue>HI8</allowedValue>
490                     <allowedValue>CD-ROM</allowedValue>
491                     <allowedValue>CD-DA</allowedValue>
492                     <allowedValue>CD-R</allowedValue>
493                     <allowedValue>CD-RW</allowedValue>
494                     <allowedValue>VIDEO-CD</allowedValue>
```

```
495                     <allowedValue>SACD</allowedValue>
496                     <allowedValue>MD-AUDIO</allowedValue>
497                     <allowedValue>MD-PICTURE</allowedValue>
498                     <allowedValue>DVD-ROM</allowedValue>
499                     <allowedValue>DVD-VIDEO</allowedValue>
500                     <allowedValue>DVD-R</allowedValue>
501                     <allowedValue>DVD+RW</allowedValue>
502                     <allowedValue>DVD-RW</allowedValue>
503                     <allowedValue>DVD-RAM</allowedValue>
504                     <allowedValue>DVD-AUDIO</allowedValue>
505                     <allowedValue>DAT</allowedValue>
506                     <allowedValue>LD</allowedValue>
507                     <allowedValue>HDD</allowedValue>
508                     <allowedValue>MICRO-MV</allowedValue>
509                     <allowedValue>NETWORK</allowedValue>
510                     <allowedValue>NONE</allowedValue>
511                     <allowedValue>NOT_IMPLEMENTED</allowedValue>
512                     <allowedValue> vendor-defined </allowedValue>
513                 </allowedValueList>
514             </stateVariable>
515             <stateVariable sendEvents="no">
516                 <name>RecordMediumWriteStatus</name>
517                 <dataType>string</dataType>
518                 <allowedValueList>
519                     <allowedValue>WRITABLE</allowedValue>
520                     <allowedValue>PROTECTED</allowedValue>
521                     <allowedValue>NOT_WRITABLE</allowedValue>
522                     <allowedValue>UNKNOWN</allowedValue>
523                     <allowedValue>NOT_IMPLEMENTED</allowedValue>
524                 </allowedValueList>
525             </stateVariable>
526             <stateVariable sendEvents="yes">
527                 <name>LastChange</name>
528                 <dataType>string</dataType>
529             </stateVariable>
530             <stateVariable sendEvents="no">
531                 <name>CurrentTrackDuration</name>
532                 <dataType>string</dataType>
533             </stateVariable>
534             <stateVariable sendEvents="no">
535                 <name>TransportState</name>
536                 <dataType>string</dataType>
537                 <allowedValueList>
538                     <allowedValue>STOPPED</allowedValue>
539                     <allowedValue>PLAYING</allowedValue>
540                     <allowedValue>TRANSITIONING</allowedValue>
541                     <allowedValue>PAUSED_PLAYBACK</allowedValue>
```

```
                    <allowedValue>PAUSED_RECORDING</allowedValue>
                    <allowedValue>RECORDING</allowedValue>
                    <allowedValue>NO_MEDIA_PRESENT</allowedValue>
                </allowedValueList>
            </stateVariable>
            <stateVariable sendEvents="no">
                <name>PossibleRecordQualityModes</name>
                <dataType>string</dataType>
            </stateVariable>
        </serviceStateTable>
    </scpd>
    <site>
        <name>Google</name>
        <url>www.google.com</url>
    </site>
    <site>
        <name>Facebook</name>
        <url>www.facebook.com</url>
    </site>
</sites>
```


1.6.3.3 urn:schemas-upnp-org:service:RenderingControl:1

```
GET /upnp/rendercontrolSCPD.xml HTTP/1.1
Accept-Encoding: gzip
User-Agent: Jetty/9.3.z-SNAPSHOT
Host: 192.168.50.44:49494

HTTP/1.1 200 OK
CONTENT-LENGTH: 13317
Accept-Ranges: bytes
CONTENT-TYPE: text/xml
DATE: Thu, 07 Mar 2024 07:15:37 GMT
LAST-MODIFIED: Thu, 01 Jan 1970 00:00:00 GMT
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic
CONNECTION: close
```


```
This XML file does not appear to have any style information associated with
it. The document tree is shown below.

<scpd
    xmlns="urn:schemas-upnp-org:service-1-0">
```

```
    <specVersion>
        <major>1</major>
        <minor>0</minor>
    </specVersion>
    <actionList>
        <action>
            <name>GetBlueVideoBlackLevel</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentBlueVideoBlackLevel</name>
                    <direction>out</direction>
<relatedStateVariable>BlueVideoBlackLevel</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetBlueVideoGain</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentBlueVideoGain</name>
                    <direction>out</direction>
                    <relatedStateVariable>BlueVideoGain</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetBrightness</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
```

```
                <argument>
                    <name>CurrentBrightness</name>
                    <direction>out</direction>
                    <relatedStateVariable>Brightness</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetColorTemperature</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentColorTemperature</name>
                    <direction>out</direction>
<relatedStateVariable>ColorTemperature</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetContrast</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentContrast</name>
                    <direction>out</direction>
                    <relatedStateVariable>Contrast</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetGreenVideoBlackLevel</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
```

```
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentGreenVideoBlackLevel</name>
                    <direction>out</direction>
<relatedStateVariable>GreenVideoBlackLevel</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetGreenVideoGain</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentGreenVideoGain</name>
                    <direction>out</direction>
                    <relatedStateVariable>GreenVideoGain</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetHorizontalKeystone</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentHorizontalKeystone</name>
                    <direction>out</direction>
<relatedStateVariable>HorizontalKeystone</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetLoudness</name>
            <argumentList>
```

```
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentLoudness</name>
                    <direction>out</direction>
                    <relatedStateVariable>Loudness</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetMute</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentMute</name>
                    <direction>out</direction>
                    <relatedStateVariable>Mute</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetRedVideoBlackLevel</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
```

```
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentRedVideoBlackLevel</name>
                    <direction>out</direction>
<relatedStateVariable>RedVideoBlackLevel</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetRedVideoGain</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentRedVideoGain</name>
                    <direction>out</direction>
                    <relatedStateVariable>RedVideoGain</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetSharpness</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentSharpness</name>
                    <direction>out</direction>
                    <relatedStateVariable>Sharpness</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetVerticalKeystone</name>
            <argumentList>
                <argument>
```

```
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentVerticalKeystone</name>
                    <direction>out</direction>
<relatedStateVariable>VerticalKeystone</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetVolume</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentVolume</name>
                    <direction>out</direction>
                    <relatedStateVariable>Volume</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetVolumeDB</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
```

```
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>CurrentVolume</name>
                    <direction>out</direction>
                    <relatedStateVariable>VolumeDB</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetVolumeDBRange</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>MinValue</name>
                    <direction>out</direction>
                    <relatedStateVariable>VolumeDB</relatedStateVariable>
                </argument>
                <argument>
                    <name>MaxValue</name>
                    <direction>out</direction>
                    <relatedStateVariable>VolumeDB</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>ListPresets</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
```

```
                    <name>CurrentPresetNameList</name>
                    <direction>out</direction>
                    <relatedStateVariable>PresetNameList</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>SetMute</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>DesiredMute</name>
                    <direction>in</direction>
                    <relatedStateVariable>Mute</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>SetVolume</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>DesiredVolume</name>
                    <direction>in</direction>
                    <relatedStateVariable>Volume</relatedStateVariable>
```

```
                </argument>
            </argumentList>
        </action>
        <action>
            <name>SetVolumeDB</name>
            <argumentList>
                <argument>
                    <name>InstanceID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_InstanceID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Channel</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_Channel</relatedStateVariable>
                </argument>
                <argument>
                    <name>DesiredVolume</name>
                    <direction>in</direction>
                    <relatedStateVariable>VolumeDB</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
    </actionList>
    <serviceStateTable>
        <stateVariable sendEvents="no">
            <name>GreenVideoGain</name>
            <dataType>ui2</dataType>
            <allowedValueRange>
                <minimum>0</minimum>
                <maximum>100</maximum>
                <step>1</step>
            </allowedValueRange>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>BlueVideoBlackLevel</name>
            <dataType>ui2</dataType>
            <allowedValueRange>
                <minimum>0</minimum>
                <maximum>100</maximum>
                <step>1</step>
            </allowedValueRange>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>VerticalKeystone</name>
```

```
            <dataType>i2</dataType>
            <allowedValueRange>
                <minimum>-32768</minimum>
                <maximum>32767</maximum>
                <step>1</step>
            </allowedValueRange>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>GreenVideoBlackLevel</name>
            <dataType>ui2</dataType>
            <allowedValueRange>
                <minimum>0</minimum>
                <maximum>100</maximum>
                <step>1</step>
            </allowedValueRange>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>Volume</name>
            <dataType>ui2</dataType>
            <allowedValueRange>
                <minimum>0</minimum>
                <maximum>100</maximum>
                <step>1</step>
            </allowedValueRange>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>Loudness</name>
            <dataType>boolean</dataType>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_InstanceID</name>
            <dataType>ui4</dataType>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>RedVideoGain</name>
            <dataType>ui2</dataType>
            <allowedValueRange>
                <minimum>0</minimum>
                <maximum>100</maximum>
                <step>1</step>
            </allowedValueRange>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>ColorTemperature</name>
            <dataType>ui2</dataType>
            <allowedValueRange>
                <minimum>0</minimum>
```

```
439                 <maximum>65535</maximum>
440                 <step>1</step>
441             </allowedValueRange>
442         </stateVariable>
443         <stateVariable sendEvents="no">
444             <name>Sharpness</name>
445             <dataType>ui2</dataType>
446             <allowedValueRange>
447                 <minimum>0</minimum>
448                 <maximum>100</maximum>
449                 <step>1</step>
450             </allowedValueRange>
451         </stateVariable>
452         <stateVariable sendEvents="no">
453             <name>A_ARG_TYPE_PresetName</name>
454             <dataType>string</dataType>
455             <allowedValueList>
456                 <allowedValue>FactoryDefaults</allowedValue>
457                 <allowedValue>InstallationDefaults</allowedValue>
458                 <allowedValue>Vendor defined</allowedValue>
459             </allowedValueList>
460         </stateVariable>
461         <stateVariable sendEvents="no">
462             <name>RedVideoBlackLevel</name>
463             <dataType>ui2</dataType>
464             <allowedValueRange>
465                 <minimum>0</minimum>
466                 <maximum>100</maximum>
467                 <step>1</step>
468             </allowedValueRange>
469         </stateVariable>
470         <stateVariable sendEvents="no">
471             <name>BlueVideoGain</name>
472             <dataType>ui2</dataType>
473             <allowedValueRange>
474                 <minimum>0</minimum>
475                 <maximum>100</maximum>
476                 <step>1</step>
477             </allowedValueRange>
478         </stateVariable>
479         <stateVariable sendEvents="no">
480             <name>Mute</name>
481             <dataType>boolean</dataType>
482         </stateVariable>
483         <stateVariable sendEvents="yes">
484             <name>LastChange</name>
485             <dataType>string</dataType>
```

```
486         </stateVariable>
487         <stateVariable sendEvents="no">
488             <name>A_ARG_TYPE_Channel</name>
489             <dataType>string</dataType>
490             <allowedValueList>
491                 <allowedValue>Master</allowedValue>
492                 <allowedValue>LF</allowedValue>
493                 <allowedValue>RF</allowedValue>
494             </allowedValueList>
495         </stateVariable>
496         <stateVariable sendEvents="no">
497             <name>HorizontalKeystone</name>
498             <dataType>i2</dataType>
499             <allowedValueRange>
500                 <minimum>-32768</minimum>
501                 <maximum>32767</maximum>
502                 <step>1</step>
503             </allowedValueRange>
504         </stateVariable>
505         <stateVariable sendEvents="no">
506             <name>VolumeDB</name>
507             <dataType>i2</dataType>
508             <allowedValueRange>
509                 <minimum>-15360</minimum>
510                 <maximum>0</maximum>
511             </allowedValueRange>
512         </stateVariable>
513         <stateVariable sendEvents="no">
514             <name>PresetNameList</name>
515             <dataType>string</dataType>
516         </stateVariable>
517         <stateVariable sendEvents="no">
518             <name>Contrast</name>
519             <dataType>ui2</dataType>
520             <allowedValueRange>
521                 <minimum>0</minimum>
522                 <maximum>100</maximum>
523                 <step>1</step>
524             </allowedValueRange>
525         </stateVariable>
526         <stateVariable sendEvents="no">
527             <name>Brightness</name>
528             <dataType>ui2</dataType>
529             <allowedValueRange>
530                 <minimum>0</minimum>
531                 <maximum>100</maximum>
532                 <step>1</step>
```

```
            </allowedValueRange>
        </stateVariable>
    </serviceStateTable>
</scpd>
```

1.6.3.4 urn:schemas-upnp-org:service:ConnectionManager:1

```
GET /upnp/renderconnmgrSCPD.xml HTTP/1.1
Accept-Encoding: gzip
User-Agent: Jetty/9.3.z-SNAPSHOT
Host: 192.168.50.44:49494

HTTP/1.1 200 OK
CONTENT-LENGTH: 4783
Accept-Ranges: bytes
CONTENT-TYPE: text/xml
DATE: Thu, 07 Mar 2024 07:15:37 GMT
LAST-MODIFIED: Thu, 01 Jan 1970 00:00:00 GMT
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic
CONNECTION: close
```

```
This XML file does not appear to have any style information associated with it. The document tree is shown below.

<scpd
    xmlns="urn:schemas-upnp-org:service-1-0">
    <specVersion>
        <major>1</major>
        <minor>0</minor>
    </specVersion>
    <actionList>
        <action>
            <name>GetCurrentConnectionIDs</name>
            <argumentList>
                <argument>
                    <name>ConnectionIDs</name>
                    <direction>out</direction>

<relatedStateVariable>CurrentConnectionIDs</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
```

```
20          <action>
21              <name>GetCurrentConnectionInfo</name>
22              <argumentList>
23                  <argument>
24                      <name>ConnectionID</name>
25                      <direction>in</direction>
26
    <relatedStateVariable>A_ARG_TYPE_ConnectionID</relatedStateVariable>
27                  </argument>
28                  <argument>
29                      <name>RcsID</name>
30                      <direction>out</direction>
31
    <relatedStateVariable>A_ARG_TYPE_RcsID</relatedStateVariable>
32                  </argument>
33                  <argument>
34                      <name>AVTransportID</name>
35                      <direction>out</direction>
36
    <relatedStateVariable>A_ARG_TYPE_AVTransportID</relatedStateVariable>
37                  </argument>
38                  <argument>
39                      <name>ProtocolInfo</name>
40                      <direction>out</direction>
41
    <relatedStateVariable>A_ARG_TYPE_ProtocolInfo</relatedStateVariable>
42                  </argument>
43                  <argument>
44                      <name>PeerConnectionManager</name>
45                      <direction>out</direction>
46
    <relatedStateVariable>A_ARG_TYPE_ConnectionManager</relatedStateVariable>
47                  </argument>
48                  <argument>
49                      <name>PeerConnectionID</name>
50                      <direction>out</direction>
51
    <relatedStateVariable>A_ARG_TYPE_ConnectionID</relatedStateVariable>
52                  </argument>
53                  <argument>
54                      <name>Direction</name>
55                      <direction>out</direction>
56
    <relatedStateVariable>A_ARG_TYPE_Direction</relatedStateVariable>
57                  </argument>
58                  <argument>
59                      <name>Status</name>
```

```
                    <direction>out</direction>
<relatedStateVariable>A_ARG_TYPE_ConnectionStatus</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>GetProtocolInfo</name>
            <argumentList>
                <argument>
                    <name>Source</name>
                    <direction>out</direction>
<relatedStateVariable>SourceProtocolInfo</relatedStateVariable>
                </argument>
                <argument>
                    <name>Sink</name>
                    <direction>out</direction>
<relatedStateVariable>SinkProtocolInfo</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
        <action>
            <name>PrepareForConnection</name>
            <argumentList>
                <argument>
                    <name>RemoteProtocolInfo</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_ProtocolInfo</relatedStateVariable>
                </argument>
                <argument>
                    <name>PeerConnectionManager</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_ConnectionManager</relatedStateVariable>
                </argument>
                <argument>
                    <name>PeerConnectionID</name>
                    <direction>in</direction>
<relatedStateVariable>A_ARG_TYPE_ConnectionID</relatedStateVariable>
                </argument>
                <argument>
                    <name>Direction</name>
                    <direction>in</direction>
```

```
<relatedStateVariable>A_ARG_TYPE_Direction</relatedStateVariable>
                </argument>
                <argument>
                    <name>ConnectionID</name>
                    <direction>out</direction>
<relatedStateVariable>A_ARG_TYPE_ConnectionID</relatedStateVariable>
                </argument>
                <argument>
                    <name>AVTransportID</name>
                    <direction>out</direction>
<relatedStateVariable>A_ARG_TYPE_AVTransportID</relatedStateVariable>
                </argument>
                <argument>
                    <name>RcsID</name>
                    <direction>out</direction>
<relatedStateVariable>A_ARG_TYPE_RcsID</relatedStateVariable>
                </argument>
            </argumentList>
        </action>
    </actionList>
    <serviceStateTable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_ConnectionManager</name>
            <dataType>string</dataType>
        </stateVariable>
        <stateVariable sendEvents="yes">
            <name>SinkProtocolInfo</name>
            <dataType>string</dataType>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_ConnectionStatus</name>
            <dataType>string</dataType>
            <allowedValueList>
                <allowedValue>OK</allowedValue>
                <allowedValue>ContentFormatMismatch</allowedValue>
                <allowedValue>InsufficientBandwidth</allowedValue>
                <allowedValue>UnreliableChannel</allowedValue>
                <allowedValue>Unknown</allowedValue>
            </allowedValueList>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_AVTransportID</name>
            <dataType>i4</dataType>
```

```
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_Direction</name>
            <dataType>string</dataType>
            <allowedValueList>
                <allowedValue>Input</allowedValue>
                <allowedValue>Output</allowedValue>
            </allowedValueList>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_RcsID</name>
            <dataType>i4</dataType>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_ProtocolInfo</name>
            <dataType>string</dataType>
        </stateVariable>
        <stateVariable sendEvents="no">
            <name>A_ARG_TYPE_ConnectionID</name>
            <dataType>i4</dataType>
        </stateVariable>
        <stateVariable sendEvents="yes">
            <name>SourceProtocolInfo</name>
            <dataType>string</dataType>
        </stateVariable>
        <stateVariable sendEvents="yes">
            <name>CurrentConnectionIDs</name>
            <dataType>string</dataType>
        </stateVariable>
    </serviceStateTable>
</scpd>
```

6.4 控制

在这个阶段中，使用的就是SOAP协议（简单对象访问协议），这SOAP基于TCP/IP协议传输HTTP的POST和M-POST命令，SOAP使用XML来说明采取的活动。

a. SetAVTransportURI

```
POST /upnp/control/rendertransport1 HTTP/1.1
Content-Type: text/xml; charset="utf-8"
HOST: 192.168.50.44
Content-Length: 1679
SOAPACTION: "urn:schemas-upnp-org:service:AVTransport:1#SetAVTransportURI"
Connection: close

```

```
<?xml version="1.0" encoding="utf-8"?>
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <s:Body>
      <u:SetAVTransportURI xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
         <InstanceID>0</InstanceID>
         <CurrentURI>http://192.168.50.84:61323/mp3?filepath=%2Fdata%2Fuser%2F0%2Fcom.netease.cloudmusic%2Ffiles%2F%E4%B8%B4%E6%97%B6%E6%96%87%E4%BB%B6&amp;decode=false</CurrentURI>
         <CurrentURIMetaData>&lt;DIDL-Lite xmlns:dc=&quot;http://purl.org/dc/elements/1.1/&quot; xmlns:upnp=&quot;urn:schemas-upnp-org:metadata-1-0/upnp/&quot; xmlns=&quot;urn:schemas-upnp-org:metadata-1-0/DIDL-Lite/&quot; xmlns:netease=&quot;http://music.163.com/dlna/&quot;&gt;&lt;item id=&quot;-1709802975619&quot;&gt;&lt;dc:title&gt;system-ready.m4a&lt;/dc:title&gt;&lt;dc:creator&gt;&amp;#26410;&amp;#30693;&lt;/dc:creator&gt;&lt;upnp:class&gt;object.item.audioItem.musicTrack&lt;/upnp:class&gt;&lt;res protocolInfo=&quot;http-get:*:audio/mpeg:DLNA.ORG_PN=MP3;DLNA.ORG_OP=01;&quot; duration=&quot;00:00:01.000&quot;&gt;http://192.168.50.84:61323/mp3?filepath=%2Fdata%2Fuser%2F0%2Fcom.netease.cloudmusic%2Ffiles%2F%E4%B8%B4%E6%97%B6%E6%96%87%E4%BB%B6&amp;amp;decode=false&lt;/res&gt;&lt;upnp:artist&gt;&amp;#26410;&amp;#30693;&lt;/upnp:artist&gt;&lt;upnp:album&gt;&amp;#26410;&amp;#30693;&amp;#19987;&amp;#36753;&lt;/upnp:album&gt;&lt;upnp:albumArtURI&gt;&lt;/upnp:albumArtURI&gt;&lt;netease:musicId&gt;-1709802975619&lt;/netease:musicId&gt;&lt;neteasemusicdlna&gt;1&lt;/neteasemusicdlna&gt;&lt;/item&gt;&lt;/DIDL-Lite&gt;</CurrentURIMetaData>
      </u:SetAVTransportURI>
   </s:Body>
</s:Envelope>
HTTP/1.1 200 OK
CONTENT-LENGTH: 270
Accept-Ranges: bytes
CONTENT-TYPE: text/xml; charset="utf-8"
DATE: Thu, 07 Mar 2024 09:18:42 GMT
EXT:
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"><s:Body>
<u:SetAVTransportURIResponse xmlns:u="urn:schemas-upnp-org:service:AVTransport:1"></u:SetAVTransportURIResponse>
</s:Body> </s:Envelope>
```

b. 播放

i. ControlPoint 控制播放

```
POST /upnp/control/rendertransport1 HTTP/1.1
Content-Type: text/xml; charset="utf-8"
HOST: 192.168.50.44
Content-Length: 349
SOAPACTION: "urn:schemas-upnp-org:service:AVTransport:1#Play"
Connection: close

<?xml version="1.0" encoding="utf-8"?>
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <s:Body>
      <u:Play xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
         <InstanceID>0</InstanceID>
         <Speed>1</Speed>
      </u:Play>
   </s:Body>
</s:Envelope>
HTTP/1.1 200 OK
CONTENT-LENGTH: 244
Accept-Ranges: bytes
CONTENT-TYPE: text/xml; charset="utf-8"
DATE: Thu, 07 Mar 2024 10:21:49 GMT
EXT:
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"><s:Body>
<u:PlayResponse xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
</u:PlayResponse>
</s:Body> </s:Envelope>
```

ii. MediaRender 响应播放

```
NOTIFY /evetSub HTTP/1.1
HOST: 192.168.50.84:8058
CONTENT-TYPE: text/xml; charset="utf-8"
CONTENT-LENGTH: 434
NT: upnp:event
NTS: upnp:propchange
```

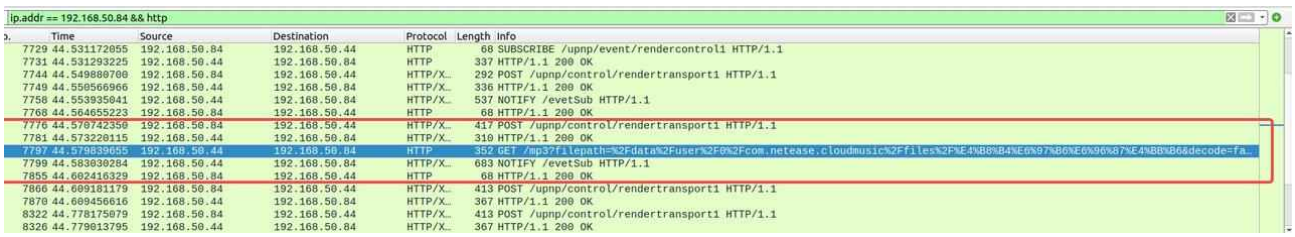
```
SID: uuid:bc37022c-dc6b-11ee-bc50-848b31c33d2c
SEQ: 7

<e:propertyset xmlns:e="urn:schemas-upnp-org:event-1-0">
<e:property>
<LastChange>&lt;?xml version="1.0"?&gt;
&lt;Event xmlns="urn:schemas-upnp-org:metadata-1-0/AVT/"&gt;
&lt;InstanceID val="0"&gt;
&lt;TransportState val="PLAYING"&gt;&lt;/TransportState&gt;
&lt;CurrentTransportActions val="PAUSE,STOP,SEEK"&gt;&lt;/CurrentTransportActions&gt;
&lt;/InstanceID&gt;
&lt;/Event&gt;
</LastChange>
</e:property>
</e:propertyset>


HTTP/1.1 200 OK
Content-Type: text/html; charset="utf-8"
Server: Linux/4.19.95 CyberHTTP/1.0
Content-Length: 0
Date: Thu, 07 Mar 2024 18:21:49 GMT
```

iii. 拉流

MediaRender 从 MediaServer拉流



No.	Time	Source	Destination	Protocol	Length	Info
7729	44.531172055	192.168.50.84	192.168.50.44	HTTP	68	SUBSCRIBE /upnp/event/rendercontrol1 HTTP/1.1
7731	44.531293225	192.168.50.44	192.168.50.84	HTTP	337	HTTP/1.1 200 OK
7744	44.549880700	192.168.50.84	192.168.50.44	HTTP/X…	292	POST /upnp/control/rendertransport1 HTTP/1.1
7749	44.550566966	192.168.50.44	192.168.50.84	HTTP/X…	336	HTTP/1.1 200 OK
7758	44.553935041	192.168.50.44	192.168.50.84	HTTP/X…	537	NOTIFY /evetSub HTTP/1.1
7768	44.564655223	192.168.50.84	192.168.50.44	HTTP	68	HTTP/1.1 200 OK
7776	44.570742350	192.168.50.84	192.168.50.44	HTTP/X…	417	POST /upnp/control/rendertransport1 HTTP/1.1
7781	44.573220115	192.168.50.44	192.168.50.84	HTTP/X…	310	HTTP/1.1 200 OK
7797	44.579039655	192.168.50.44	192.168.50.84	HTTP	353	GET /mp3?filepath=%2Fdata%2Fuser%2F0%2Fcom.netease.cloudmusic%2Ffiles%2F%E4%B8%B4%E6%97%B6%E6%96%87%E4%BB%B6&decode=fa…
7799	44.583030284	192.168.50.44	192.168.50.84	HTTP/X…	683	NOTIFY /evetSub HTTP/1.1
7855	44.602416329	192.168.50.84	192.168.50.44	HTTP	68	HTTP/1.1 200 OK
7866	44.609181179	192.168.50.84	192.168.50.44	HTTP/X…	413	POST /upnp/control/rendertransport1 HTTP/1.1
7870	44.609456616	192.168.50.44	192.168.50.84	HTTP/X…	367	HTTP/1.1 200 OK
8322	44.778175079	192.168.50.84	192.168.50.44	HTTP/X…	413	POST /upnp/control/rendertransport1 HTTP/1.1
8326	44.779013795	192.168.50.44	192.168.50.84	HTTP/X…	367	HTTP/1.1 200 OK

```
GET /mp3?filepath=%2Fdata%2Fuser%2F0%2Fcom.netease.cloudmusic%2Ffiles%2F%E4%B8%B4%E6%97%B6%E6%96%87%E4%BB%B6&decode=false HTTP/1.1
Host: 192.168.50.84:61323
User-Agent: GStreamer souphttpsrc 1.20.3 libsoup/2.74.2
icy-metadata: 1
Accept-Encoding: identity
Connection: Keep-Alive
```

c. 暂停

i. ControlPoint 控制暂停

```
POST /upnp/control/rendertransport1 HTTP/1.1
Content-Type: text/xml; charset="utf-8"
HOST: 192.168.50.44
Content-Length: 325
SOAPACTION: "urn:schemas-upnp-org:service:AVTransport:1#Pause"
Connection: close

<?xml version="1.0" encoding="utf-8"?>
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <s:Body>
      <u:Pause xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
         <InstanceID>0</InstanceID>
      </u:Pause>
   </s:Body>
</s:Envelope>
HTTP/1.1 200 OK
CONTENT-LENGTH: 246
Accept-Ranges: bytes
CONTENT-TYPE: text/xml; charset="utf-8"
DATE: Thu, 07 Mar 2024 10:16:29 GMT
EXT:
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"><s:Body>
<u:PauseResponse xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
</u:PauseResponse>
</s:Body> </s:Envelope>
```

ii. MediaRender 响应暂停

```
NOTIFY /evetSub HTTP/1.1
HOST: 192.168.50.84:8058
CONTENT-TYPE: text/xml; charset="utf-8"
CONTENT-LENGTH: 441
NT: upnp:event
NTS: upnp:propchange
SID: uuid:bc37022c-dc6b-11ee-bc50-848b31c33d2c
SEQ: 4
```

```
<e:propertyset xmlns:e="urn:schemas-upnp-org:event-1-0">
<e:property>
<LastChange>&lt;?xml version="1.0"?&gt;
&lt;Event xmlns="urn:schemas-upnp-org:metadata-1-0/AVT/"&gt;
&lt;InstanceID val="0"&gt;
&lt;TransportState val="PAUSED_PLAYBACK"&gt;&lt;/TransportState&gt;
&lt;CurrentTransportActions val="PLAY,STOP,SEEK"&gt;&lt;/CurrentTransportActions&gt;
&lt;/InstanceID&gt;
&lt;/Event&gt;
</LastChange>
</e:property>
</e:propertyset>


HTTP/1.1 200 OK
Content-Type: text/html; charset="utf-8"
Server: Linux/4.19.95 CyberHTTP/1.0
Content-Length: 0
Date: Thu, 07 Mar 2024 18:16:29 GMT


```

d. GetPositionInfo

```
POST /upnp/control/rendertransport1 HTTP/1.1
Content-Type: text/xml; charset="utf-8"
HOST: 192.168.50.44
Content-Length: 345
SOAPACTION: "urn:schemas-upnp-org:service:AVTransport:1#GetPositionInfo"
Connection: close

<?xml version="1.0" encoding="utf-8"?>
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <s:Body>
      <u:GetPositionInfo xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
         <InstanceID>0</InstanceID>
      </u:GetPositionInfo>
   </s:Body>
</s:Envelope>
HTTP/1.1 200 OK
CONTENT-LENGTH: 1756
```

```
Accept-Ranges: bytes
CONTENT-TYPE: text/xml; charset="utf-8"
DATE: Thu, 07 Mar 2024 09:18:42 GMT
EXT:
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"><s:Body>
<u:GetPositionInfoResponse xmlns:u="urn:schemas-upnp-org:service:AVTransport:1">
<Track>1</Track>
<TrackDuration>0:00:00</TrackDuration>
<TrackMetaData>&lt;DIDL-Lite xmlns:dc=&quot;http://purl.org/dc/elements/1.1/&quot; xmlns:upnp=&quot;urn:schemas-upnp-org:metadata-1-0/upnp/&quot; xmlns=&quot;urn:schemas-upnp-org:metadata-1-0/DIDL-Lite/&quot; xmlns:netease=&quot;http://music.163.com/dlna/&quot;&gt;&lt;item id=&quot;-1709802975619&quot;&gt;&lt;dc:title&gt;system-ready.m4a&lt;/dc:title&gt;&lt;dc:creator&gt;&amp;#26410;&amp;#30693;&lt;/dc:creator&gt;&lt;upnp:class&gt;object.item.audioItem.musicTrack&lt;/upnp:class&gt;&lt;res protocolInfo=&quot;http-get:*:audio/mpeg:DLNA.ORG_PN=MP3;DLNA.ORG_OP=01;&quot; duration=&quot;00:00:01.000&quot;&gt;http://192.168.50.84:61323/mp3?filepath=%2Fdata%2Fuser%2F0%2Fcom.netease.cloudmusic%2Ffiles%2F%E4%B8%B4%E6%97%B6%E6%96%87%E4%BB%B6&amp;amp;decode=false&lt;/res&gt;&lt;upnp:artist&gt;&amp;#26410;&amp;#30693;&lt;/upnp:artist&gt;&lt;upnp:album&gt;&amp;#26410;&amp;#30693;&amp;#19987;&amp;#36753;&lt;/upnp:album&gt;&lt;upnp:albumArtURI&gt;&lt;/upnp:albumArtURI&gt;&lt;netease:musicId&gt;-1709802975619&lt;/netease:musicId&gt;&lt;neteasemusicdlna&gt;1&lt;/neteasemusicdlna&gt;&lt;/item&gt;&lt;/DIDL-Lite&gt;</TrackMetaData>
<TrackURI>http://192.168.50.84:61323/mp3?filepath=%2Fdata%2Fuser%2F0%2Fcom.netease.cloudmusic%2Ffiles%2F%E4%B8%B4%E6%97%B6%E6%96%87%E4%BB%B6&amp;decode=false</TrackURI>
<RelTime>0:00:00</RelTime>
<AbsTime>NOT_IMPLEMENTED</AbsTime>
<RelCount>2147483647</RelCount>
<AbsCount>2147483647</AbsCount>
</u:GetPositionInfoResponse>
</s:Body> </s:Envelope>
```

e. GetVolume

```
POST /upnp/control/rendercontrol1 HTTP/1.1
```

```
Content-Type: text/xml; charset="utf-8"
HOST: 192.168.50.44
Content-Length: 373
SOAPACTION: "urn:schemas-upnp-org:service:RenderingControl:1#GetVolume"
Connection: close

<?xml version="1.0" encoding="utf-8"?>
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <s:Body>
      <u:GetVolume xmlns:u="urn:schemas-upnp-org:service:RenderingControl:1">
         <InstanceID>0</InstanceID>
         <Channel>Master</Channel>
      </u:GetVolume>
   </s:Body>
</s:Envelope>
HTTP/1.1 200 OK
CONTENT-LENGTH: 297
Accept-Ranges: bytes
CONTENT-TYPE: text/xml; charset="utf-8"
DATE: Thu, 07 Mar 2024 09:18:42 GMT
EXT:
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
X-User-Agent: redsonic

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"><s:Body>
<u:GetVolumeResponse xmlns:u="urn:schemas-upnp-org:service:RenderingControl:1">
<CurrentVolume>100</CurrentVolume>
</u:GetVolumeResponse>
</s:Body> </s:Envelope>
```

f. 播放结束

MediaRender通知ControlPoint

```
NOTIFY /evetSub HTTP/1.1
HOST: 192.168.50.84:8058
CONTENT-TYPE: text/xml; charset="utf-8"
CONTENT-LENGTH: 775
NT: upnp:event
NTS: upnp:propchange
SID: uuid:d41ed280-dc6a-11ee-a0d5-c2c96d1d5848
```

```
SEQ: 6

<e:propertyset xmlns:e="urn:schemas-upnp-org:event-1-0">
<e:property>
<LastChange>&lt;?xml version="1.0"?&gt;
&lt;Event xmlns="urn:schemas-upnp-org:metadata-1-0/AVT/"&gt;
&lt;InstanceID val="0"&gt;
&lt;AVTransportURI val=""&gt;&lt;/AVTransportURI&gt;
&lt;AVTransportURIMetaData val=""&gt;&lt;/AVTransportURIMetaData&gt;
&lt;NumberOfTracks val="0"&gt;&lt;/NumberOfTracks&gt;
&lt;CurrentTrack val="0"&gt;&lt;/CurrentTrack&gt;
&lt;CurrentTrackURI val=""&gt;&lt;/CurrentTrackURI&gt;
&lt;CurrentTrackMetaData val=""&gt;&lt;/CurrentTrackMetaData&gt;
&lt;TransportState val="STOPPED"&gt;&lt;/TransportState&gt;
&lt;CurrentTransportActions val="PLAY"&gt;&lt;/CurrentTransportActions&gt;
&lt;/InstanceID&gt;
&lt;/Event&gt;
</LastChange>
</e:property>
</e:propertyset>


HTTP/1.1 200 OK
Content-Type: text/html; charset="utf-8"
Server: Linux/4.19.95 CyberHTTP/1.0
Content-Length: 0
Date: Thu, 07 Mar 2024 18:11:50 GMT
```

6.5 事件

描述阶段结束后,ControlPoint与MediaRender建立连接后,紧接着就是AVTransport Service和RenderingControl Service中Eventing的订阅逻辑交互。在服务进行的整个时间段内，ControlPoint只要变量值发生了变化或者是模式的状态发生了改变，就产生了一个事件,并将事件发布到MediaRender。

a. ControlPoint向MediaRender设备请求事件订阅消息

```
SUBSCRIBE /upnp/event/rendertransport1 HTTP/1.1
Content-Length: 0
HOST: 192.168.50.44
SID: uuid:ff7ff784-dc5a-11ee-8c03-e48a1e9266a6
TIMEOUT: Second-36000
Connection: close

HTTP/1.1 200 OK
```

```
DATE: Thu, 07 Mar 2024 08:20:19 GMT
SERVER: Linux/6.5.0-21-generic, UPnP/1.0, Portable SDK for UPnP devices/1.8.4
CONTENT-LENGTH: 0
Accept-Ranges: bytes
X-User-Agent: redsonic
SID: uuid:ff7ff784-dc5a-11ee-8c03-e48a1e9266a6
TIMEOUT: Second-36000

```

返回状态信息

```
NOTIFY /evetSub HTTP/1.1
HOST: 192.168.50.84:8058
CONTENT-TYPE: text/xml; charset="utf-8"
CONTENT-LENGTH: 2168
NT: upnp:event
NTS: upnp:propchange
SID: uuid:a3767d6a-dc5e-11ee-9188-83a86abc1ab4
SEQ: 0

<e:propertyset xmlns:e="urn:schemas-upnp-org:event-1-0">
<e:property>
<LastChange>&lt;?xml version="1.0"?&gt;
&lt;Event xmlns="urn:schemas-upnp-org:metadata-1-0/AVT/"&gt;
&lt;InstanceID val="0"&gt;
&lt;TransportStatus val="OK"&gt;&lt;/TransportStatus&gt;
&lt;NextAVTransportURI val=""&gt;&lt;/NextAVTransportURI&gt;
&lt;NextAVTransportURIMetaData val=""&gt;&lt;/NextAVTransportURIMetaData&gt;
&lt;CurrentTrackMetaData val=""&gt;&lt;/CurrentTrackMetaData&gt;
&lt;RelativeCounterPosition val="2147483647"&gt;&lt;/RelativeCounterPosition&gt;
&lt;PlaybackStorageMedium val="UNKNOWN"&gt;&lt;/PlaybackStorageMedium&gt;
&lt;RelativeTimePosition val="0:00:00"&gt;&lt;/RelativeTimePosition&gt;
&lt;PossibleRecordStorageMedia val="NOT_IMPLEMENTED"&gt;&lt;/PossibleRecordStorageMedia&gt;
&lt;CurrentPlayMode val="NORMAL"&gt;&lt;/CurrentPlayMode&gt;
&lt;TransportPlaySpeed val="1"&gt;&lt;/TransportPlaySpeed&gt;
&lt;PossiblePlaybackStorageMedia val="NETWORK,UNKNOWN"&gt;&lt;/PossiblePlaybackStorageMedia&gt;
&lt;AbsoluteTimePosition val="NOT_IMPLEMENTED"&gt;&lt;/AbsoluteTimePosition&gt;
&lt;CurrentTrack val="0"&gt;&lt;/CurrentTrack&gt;
&lt;CurrentTrackURI val=""&gt;&lt;/CurrentTrackURI&gt;
&lt;CurrentTransportActions val="PLAY"&gt;&lt;/CurrentTransportActions&gt;
```

```
&lt;NumberOfTracks val="0"&gt;&lt;/NumberOfTracks&gt;
&lt;AVTransportURI val=""&gt;&lt;/AVTransportURI&gt;
&lt;AbsoluteCounterPosition val="2147483647"&gt;&lt;/AbsoluteCounterPosition&gt;
&lt;CurrentRecordQualityMode val="NOT_IMPLEMENTED"&gt;&lt;/CurrentRecordQualityMode&gt;
&lt;CurrentMediaDuration val=""&gt;&lt;/CurrentMediaDuration&gt;
&lt;AVTransportURIMetaData val=""&gt;&lt;/AVTransportURIMetaData&gt;
&lt;RecordStorageMedium val="NOT_IMPLEMENTED"&gt;&lt;/RecordStorageMedium&gt;
&lt;RecordMediumWriteStatus val="NOT_IMPLEMENTED"&gt;&lt;/RecordMediumWriteStatus&gt;
&lt;CurrentTrackDuration val="0:00:00"&gt;&lt;/CurrentTrackDuration&gt;
&lt;TransportState val="STOPPED"&gt;&lt;/TransportState&gt;
&lt;PossibleRecordQualityModes val="NOT_IMPLEMENTED"&gt;&lt;/PossibleRecordQualityModes&gt;
&lt;/InstanceID&gt;
&lt;/Event&gt;
</LastChange>
</e:property>
</e:propertyset>


HTTP/1.1 200 OK
Content-Type: text/html; charset="utf-8"
Server: Linux/4.19.95 CyberHTTP/1.0
Content-Length: 0
Date: Thu, 07 Mar 2024 16:42:31 GMT


```

a. 推送暂停事件

```
NOTIFY /evetSub HTTP/1.1
HOST: 192.168.50.84:8058
CONTENT-TYPE: text/xml; charset="utf-8"
CONTENT-LENGTH: 441
NT: upnp:event  #事件类型
NTS: upnp:propchange
SID: uuid:112ec778-dc5e-11ee-8431-bb6176f79da6
SEQ: 4

<e:propertyset xmlns:e="urn:schemas-upnp-org:event-1-0">
<e:property>
<LastChange>&lt;?xml version="1.0"?&gt;
```

```
&lt;Event xmlns="urn:schemas-upnp-org:metadata-1-0/AVT/"&gt;
&lt;InstanceID val="0"&gt;
&lt;TransportState val="PAUSED_PLAYBACK"&gt;&lt;/TransportState&gt;  #暂停
&lt;CurrentTransportActions val="PLAY,STOP,SEEK"&gt;&lt;/CurrentTransportActions&gt;
&lt;/InstanceID&gt;
&lt;/Event&gt;
</LastChange>
</e:property>
</e:propertyset>


HTTP/1.1 200 OK
Content-Type: text/html; charset="utf-8"
Server: Linux/4.19.95 CyberHTTP/1.0
Content-Length: 0
Date: Thu, 07 Mar 2024 16:38:46 GMT

```

四，gmediarender

1. gmediarender 是一个开源的 UPnP 媒体渲染器软件，用于将音频流媒体从 DLNA/UPnP 服务器（如 NAS、电脑等）传输到支持 UPnP 协议的音频设备（如智能音箱、网络音响等）。下面是对 gmediarender 的详细解释：
   a. UPnP 媒体渲染器：UPnP（Universal Plug and Play）是一种网络协议，用于设备之间的互联，允许设备自动发现彼此并共享多媒体内容。UPnP 媒体渲染器是能够接收、解码和播放来自 UPnP 服务器的音频内容的设备。
   b. gmediarender 的功能：
      - 音频传输：gmediarender 可以从 UPnP 服务器接收音频流，并将其传输到支持 UPnP 协议的音频设备。
      - WAV 和 MP3 支持：gmediarender 支持播放 WAV 和 MP3 格式的音频文件。
      - 资源友好：gmediarender 是一个轻量级程序，资源占用较少，适合在资源受限的设备上运行。
      - 运行在 Linux 系统上：gmediarender 是为 Linux 系统设计的 UPnP 媒体渲染器软件。
2. 启动命令
   a. 控制描述：指用于提供 gmediarender 服务描述和控制信息的端口。通常情况下，这个端口是使用 UPnP（Universal Plug and Play）协议的设备上的一个端口，用于描述该设备的功能和允许其他设备与之交互。

在大多数情况下，gmediarender 使用的控制描述端口是 49494。这个端口用于提供 UPnP 服务描述文件和控制接口，允许其他设备发现和与 gmediarender 进行交互，比如开始、停止、暂停或改变播放状态等。

如果需要访问 gmediarender 控制描述端口，可以通过设备的 IP 地址和端口号来访问，例如：`http://device_ip:49494/description.xml`。这将返回 gmediarender 的服务描述文件，包含该设备的详细信息和支持的控制操作。

需要注意的是，具体的端口号可能因设备不同、配置不同而有所不同，但默认情况下 gmediarender 的控制描述端口是 49494。

```
# /usr/bin/gmediarender        可执行程序名称
# -f "Cosmos-developer"        设备名称
# --logfile=/dev/stdout        日志路径
# -p 49494                     指定控制描述端口
# -I 127.0.0.1                 设置控制描述host(默认局域网ip)

/usr/bin/gmediarender -f "Cosmos-developer" --logfile=/dev/stdout -p 49496 -I 127.0.0.1
```

3. 端口监听
   a. 192.168.50.158:49494： 控制描述host:port
   b. 0.0.0.0:1900：设备在 UPnP 网络中使用端口 1900 来广播他们的存在，以便其他设备能够发现并与他们进行通信。
   c. 端口监听情况

```
tcp        0      0 192.168.50.158:49494    0.0.0.0:*               LISTEN
      7305/gmediarender        # 监听连接
udp        0      0 0.0.0.0:1900            0.0.0.0:*
      7305/gmediarender        # 发现
```

4. 组播通知消息
   a. 组播地址
      i. 239.255.255.250

```
06:25:56.425832 IP 192.168.50.129.8008 > 239.255.255.250.1900: UDP, length 101
06:25:57.057965 IP 192.168.50.129.8008 > 239.255.255.250.1900: UDP, length 101
06:25:57.499492 IP 192.168.50.129.8008 > 239.255.255.250.1900: UDP, length 101
```

      ii. loopback

    1. 不支持

5. 描述服务

    a. 音量控制

```
06:28:18.992754 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [.], ack 1, win 128, options [nop,nop,TS val 2348173540 ecr 274503], length 0
06:28:18.993523 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [P.], seq 1:221, ack 1, win 128, options [nop,nop,TS val 2348173541 ecr 274503], length 220
06:28:18.993566 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 221, win 235, options [nop,nop,TS val 274503 ecr 2348173541], length 0
06:28:18.996224 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [P.], seq 221:639, ack 1, win 128, options [nop,nop,TS val 2348173544 ecr 274503], length 4
18
06:28:18.996263 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], length 0
06:28:18.997132 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [P.], seq 1:251, ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], length 2
50
06:28:18.997203 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [FP.], seq 251:510, ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], lengt
h 259
06:28:19.003803 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [.], ack 251, win 131, options [nop,nop,TS val 2348173552 ecr 274504], length 0
06:28:19.004860 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [F.], seq 639, ack 511, win 133, options [nop,nop,TS val 2348173553 ecr 274504], length 0
06:28:19.004890 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 640, win 243, options [nop,nop,TS val 274506 ecr 2348173553], length 0
^X^C
```

    b. 暂停/播放

```
06:28:18.992754 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [.], ack 1, win 128, options [nop,nop,TS val 2348173540 ecr 274503], length 0
06:28:18.993523 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [P.], seq 1:221, ack 1, win 128, options [nop,nop,TS val 2348173541 ecr 274503], length 220
06:28:18.993566 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 221, win 235, options [nop,nop,TS val 274503 ecr 2348173541], length 0
06:28:18.996224 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [P.], seq 221:639, ack 1, win 128, options [nop,nop,TS val 2348173544 ecr 274503], length 4
18
06:28:18.996263 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], length 0
06:28:18.997132 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [P.], seq 1:251, ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], length 2
50
06:28:18.997203 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [FP.], seq 251:510, ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], lengt
h 259
06:28:19.003803 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [.], ack 251, win 131, options [nop,nop,TS val 2348173552 ecr 274504], length 0
06:28:19.004860 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [F.], seq 639, ack 511, win 133, options [nop,nop,TS val 2348173553 ecr 274504], length 0
06:28:19.004890 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 640, win 243, options [nop,nop,TS val 274506 ecr 2348173553], length 0
^X^C
```

    c. 设置音频

```
06:28:18.992754 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [.], ack 1, win 128, options [nop,nop,TS val 2348173540 ecr 274503], length 0
06:28:18.993523 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [P.], seq 1:221, ack 1, win 128, options [nop,nop,TS val 2348173541 ecr 274503], length 220
06:28:18.993566 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 221, win 235, options [nop,nop,TS val 274503 ecr 2348173541], length 0
06:28:18.996224 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [P.], seq 221:639, ack 1, win 128, options [nop,nop,TS val 2348173544 ecr 274503], length 4
18
06:28:18.996263 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], length 0
06:28:18.997132 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [P.], seq 1:251, ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], length 2
50
06:28:18.997203 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [FP.], seq 251:510, ack 639, win 243, options [nop,nop,TS val 274504 ecr 2348173544], lengt
h 259
06:28:19.003803 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [.], ack 251, win 131, options [nop,nop,TS val 2348173552 ecr 274504], length 0
06:28:19.004860 IP 192.168.50.129.46892 > 192.168.50.158.49494: Flags [F.], seq 639, ack 511, win 133, options [nop,nop,TS val 2348173553 ecr 274504], length 0
06:28:19.004890 IP 192.168.50.158.49494 > 192.168.50.129.46892: Flags [.], ack 640, win 243, options [nop,nop,TS val 274506 ecr 2348173553], length 0
^X^C
```

五、机器人局域网测试验证

1. 将android，机器连接在同一路由器下
2. 登录开服server

```
# 开启监听服务器
# 0.0.0.0     当前ip
# developer   服务名称
# stdout      输出日志到控制台
gmediarender -I 0.0.0.0 -f "developer" --logfile=stdout
```

3. android 推流并设置音量

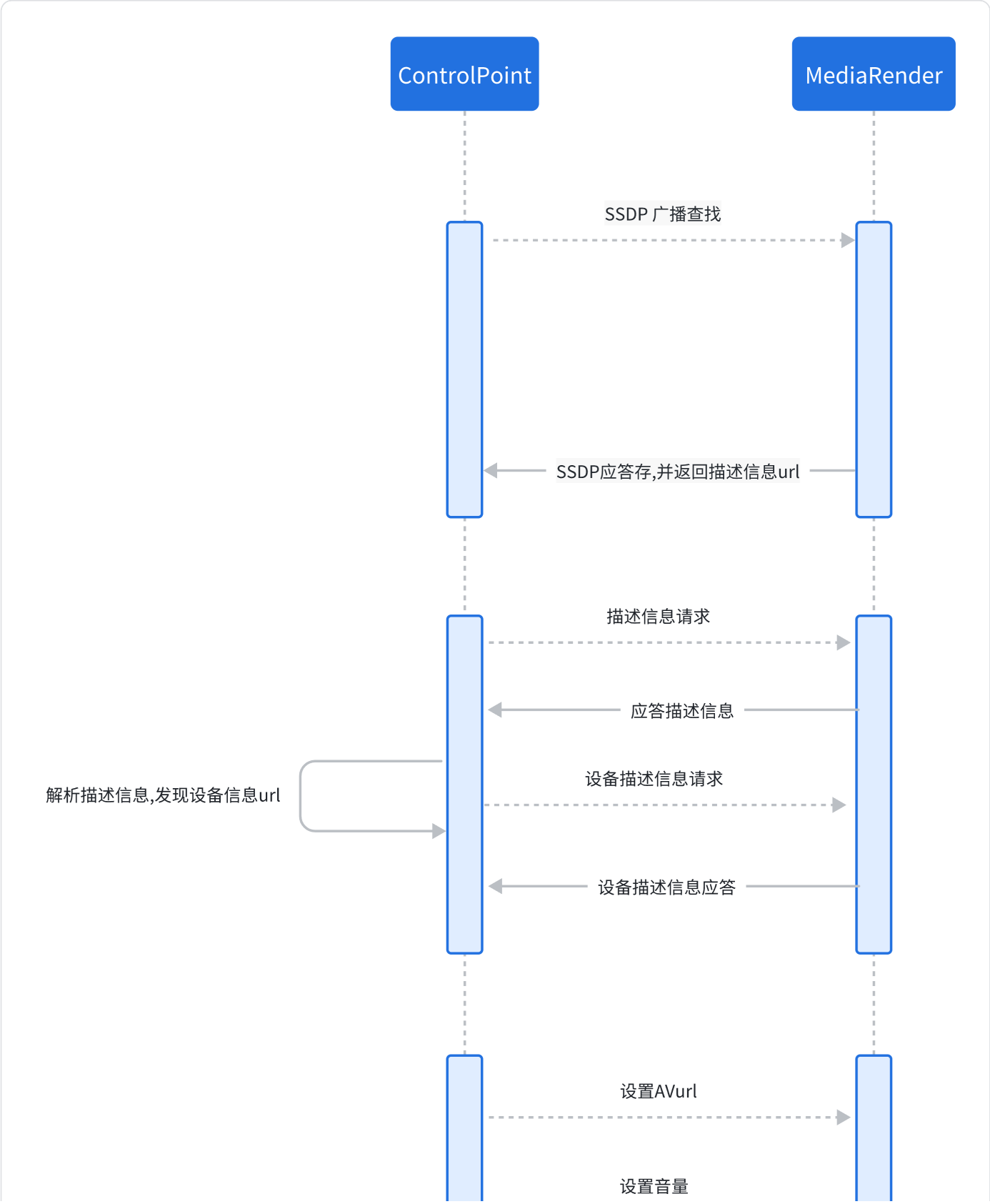
    a. 打开网易云

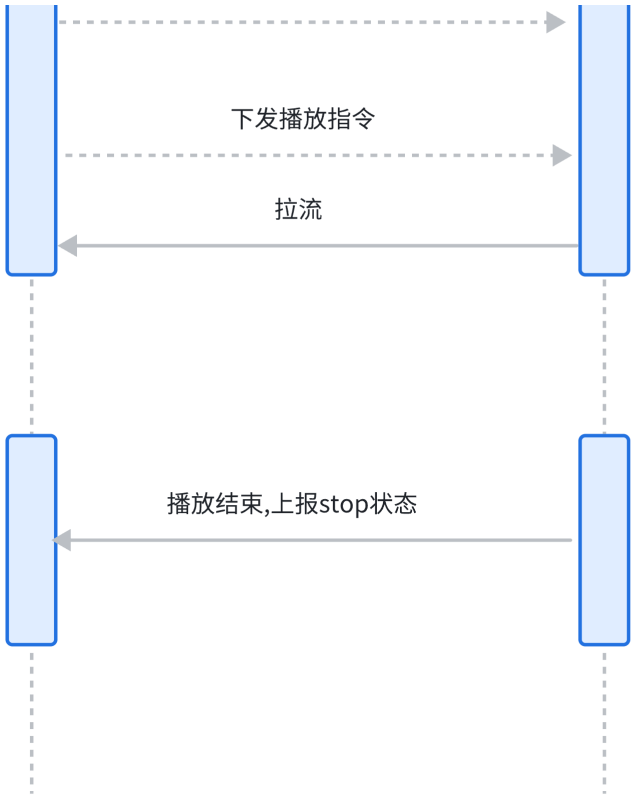
    b. 播放音乐

c. 选择投屏(默认只支持.wav,.m4a 格式，其他格式须在下位机安装额外gstreamer插件)

八、时序图

1. 局域网测试
   a. android : ControlPoint
   b. (gmediarender) : MediaRender




下发播放指令
拉流
播放结束,上报stop状态